

Immersive Analytics for Ergonomics Evaluation in Virtual Reality

Simon Kloiber*

Graz University of Technology, CGV

Reinhold Preiner

Graz University of Technology, CGV

Nicole Weidinger

Graz University of Technology, CGV

Katharina Krösl

VRVis Forschungs-GmbH

Eva Eggeling

Fraunhofer Austria GmbH

Tobias Schreck

Graz University of Technology, CGV

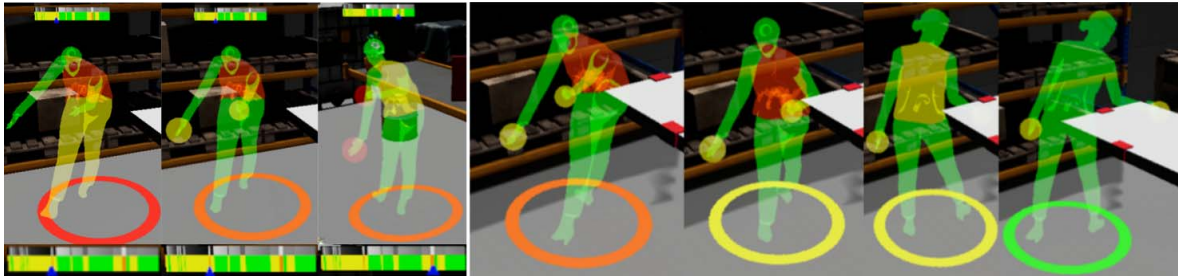


Figure 1: We introduce an immersive ergonomics evaluation system that employs avatars with color-coded body parts to show ergonomic properties of users' tracked motion. **Left:** Pose filter display for high-risk poses. **Right:** Key-frames of session replay.

ABSTRACT

Utilizing the immersiveness and spatial awareness of virtual reality, we aim to improve ergonomic evaluations, which are key to avoid musculoskeletal disorders. We present the first iteration of an immersive analytics system for ergonomic evaluation and analysis, providing real-time and offline feedback in the same virtual environment. In addition, we provide insights from an expert interview confirming the validity and practicability of our approach.

Index Terms: Human-centered computing—Visualization systems and tools; Human-centered computing—Virtual reality; Applied computing—Life and medical sciences

1 INTRODUCTION

Human labor is indispensable in many work processes, but humans are prone to developing musculoskeletal disorders (MSDs) as a result of overloading physical work [3]. MSDs are the most common labor medical problems among workers [7] and affect almost all occupations and sectors. They lead to discomfort, pain, recurring injuries and the inability to work as well as economic consequences for businesses due to productivity losses, compensation and replacement of personnel [3, 7]. Thus, the prevention of MSDs is of great interest to industry and society. It can be achieved by considering ergonomic factors in workplace design and sequences by employing one of the many different ergonomic evaluation methods. Still, this can be a complex and time-consuming task.

Most matured and accepted approaches simplify ergonomic analysis via the simulation of humans in the workplace [10]. Being purely simulated, the subjective and interactive experience in the real-world workplace may differ from the objective result obtained by the simulation [15]. Immersive virtual environments (VEs), like virtual reality (VR), combine the advantages of virtual and physical representations: money and time can be saved thanks to a virtual model of the workplace where we can measure aspects of user movement and behavior, while users can still experience it more realistically than by just observing it on a 2D computer screen. Previous research

has shown that immersive assessment of ergonomics is suitable for assessment of visibility, climbing, postures, space, reach and use of tools [2]. While such systems have not yet gained the same maturity and acceptance as classic simulation methods, there is a benefit in analyzing and exploiting their potential as immersive technologies become more popular and accessible. With this technology also comes a wealth of tracking methods and the necessity to evaluate their applicability to ergonomics evaluation.

We see a distinct lack of ergonomics evaluation systems that enable users to simulate and capture their movement in a virtual environment, and to then analyze the ergonomics of their movement in the same virtual environment. This raises the following three research questions that we seek to answer:

- *Q1: How can we use motion data of users, tracked in a VE, to assess ergonomic aspects?*
- *Q2: How can we apply immersive analytics methods for ergonomic analysis?*
- *Q3: Can we use our immersive ergonomics analysis system to compare tracking and ergonomics evaluation methods?*

The goal of our work is a step towards an immersive ergonomics analysis approach with real-time feedback and analytics directly in the VE the data was captured in. This provides valuable spatial context and thus promises to improve understanding and the efficiency of the evaluation processes. It also has potential to reduce the need for assistance, as users can better comprehend the results without depending on experts to perform an interpretation.

This work contributes a *comprehensive ergonomics evaluation and analytics system in VR with a low-cost tracking system* that allows users to simulate ergonomic tasks and motion and *analyze the ergonomic aspects of the resulting motion data in the same VE it was captured in*. The system offers both real-time feedback on the ergonomic assessment and an immersive analytics workflow to evaluate the performed motion, e.g., for workplace designs or therapy sessions. We conducted a *qualitative evaluation* of our approach based on an expert interview with a physiotherapist and discuss the current state and future of the system.

2 IMMERSIVE ANALYTICS ERGONOMICS SYSTEM

The intended immersive analysis aims to first, give users a good first impression of the session and its ergonomic quality, second,

*s.kloiber@cgv.tugraz.at

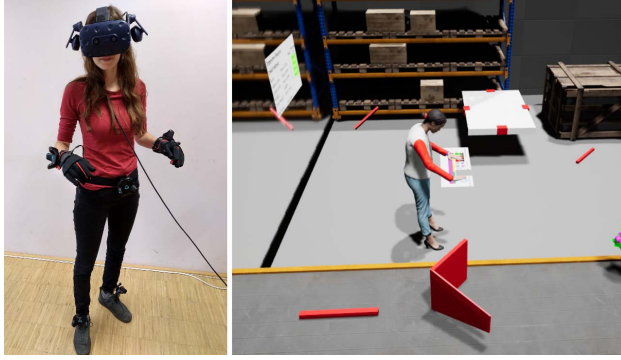


Figure 2: Left: real tracker setup with five trackers and the head mounted display. Right: sample scene showing a user interacting with our analysis panel.

to convey comprehensibly where, when and how often ergonomic problems were identified during the session and third, to provide information about why postures are rated as problematic. The system was implemented in the Unreal Engine [4], allowing users to quickly adapt their designs in the engine and to re-evaluate them on-the-fly.

2.1 Virtual Avatar

To accurately measure the ergonomics of a user, we need an accurate representation of a user's real pose by tracking the user's movement. To balance a laborious setup of the tracking system and the cost of acquiring it, our approach currently uses five trackers that are strapped to the body and can be tracked by the VR system. We focus on the use of consumer grade hardware, but have left the system flexible enough to allow switching to an improved tracking system in the future. Besides its necessity for the ergonomic evaluation, our full-body virtual avatar also increases presence for users [11]. To further improve intuitiveness and to provide natural interaction, we employ Manus VR DK2 [14] gloves that track finger movement.

The tracking is performed with an HTC Vive and five of its trackers. These trackers can be strapped to the body easily and provide precise tracking, as is necessary for VR devices. The trackers are placed on all limbs and on the pelvis (Figure 2). The pelvis determines the pivot position and main rotation of the avatar. We use Unreal Engine's skeleton animation and inverse kinematic (IK) features to calculate the full skeleton of our avatar based on the trackers' orientations and rotations. We use the IK solver FABRIK [1] for positioning the spine and head and use the engine's two-bone IK solver for positioning all limbs. The pelvis is positioned such that the FABRIK IK along the spine can position the head at the location and rotation tracked by the head mounted display. Since the tracker IDs may change between uses of the system and different users have different body measures, the first step that needs to be performed in our VE is a calibration of the avatar. This calibration registers the correspondences between the trackers and represented body parts and scales the avatar to fit the user's size.

2.2 Ergonomic Evaluation Method

The International Ergonomics Association [8] classifies aspects of work that involve human activity into three ergonomic perspectives: physical, cognitive, and organizational ergonomics. In general, all three are regarded as equally important and all of them should be considered for the design of inclusive workplaces [3].

In this work, we focus on the physical side of ergonomics. To evaluate physical workload, many different methods have been developed. Currently, we use the Ovako Working Posture Analysing System (OWAS), a widely used method for MSD evaluation, which assesses the posture of the whole body in specific time intervals.

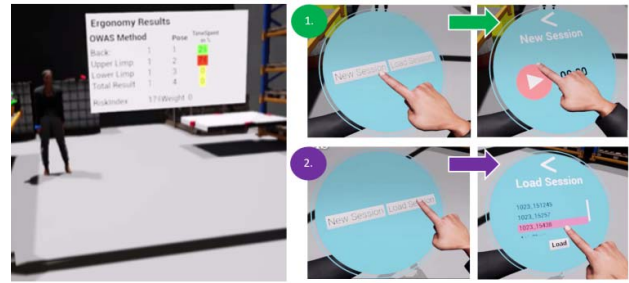


Figure 3: Left: real-time ergonomic feedback for current pose. Right: watch menu that opens when being looked at.

OWAS considers the position of the back, the arms, the legs, and the weight of lifted objects [12]. Each body part and the weight of the lifted object (defined in-engine) is assigned a grade, which results in a four-digit code, which is then converted to one of four body posture classes. We found two additions of OWAS to be important for our evaluation. First, an additional time-related risk evaluation where the time that is spent in each OWAS classes is assessed and categorized into three risk levels [5]. Second, we use the risk or Lundqvist index I [13], which represents the overall quality of the session as one number. This can be helpful to get a first impression of the quality and helps to make different OWAS analyzes easily comparable [6], since it is straightforward to check whether ergonomic improvements were successful by looking at the index.

The given motion-capturing resources do not allow a precise determination of the torso's twist. Therefore, the last two categories for the back are adapted, similar to Iqbal et al. [9]. In practice, the motion range in the real world is potentially limited during a VR session—requiring the user to teleport several times to move around in the virtual world. Instead of determining a twist of the torso, we check if the torso is bent to the side. In addition, the motion range in the real world may be limited, requiring users to teleport as a mode of transport in the virtual world. In such a case, the sixth category for the legs (body moved by legs) is also registered by detecting unnaturally big changes in the pelvis position.

2.3 Interaction

We make use of the Manus VR gloves to provide gestural interactions and provide user interfaces that resemble the real world. To keep nausea at a minimum, we use physical walking and teleportation as locomotion techniques. We implemented three gestures: grabbing for performing actions in the world, a finger gun to start teleportation and subsequent lowering of the thumb to perform the teleport. The first finger gun gesture starts the avatar calibration, all following ones will start the teleportation process. We have created two user interfaces in our system: a wrist watch menu (Figure 3 right), which appears when the user looks at their wrist, and an analysis panel (Figure 4) displayed at pelvis height.

2.4 Visualization

To enable users to detect possible ergonomic problems early on, OWAS evaluation results are continuously calculated and presented in real-time via an analysis panel that is positioned at a central position in the scene (Figure 3 (left)). Evaluation sessions can be saved and contain avatar calibration, all tracking information and the OWAS evaluation results. Once a user loads a previous session, the analysis panel appears.

2.4.1 Analysis Panel

The analysis panel displayed in Figure 4 is the main interaction point during the offline analysis of an evaluation session. It displays

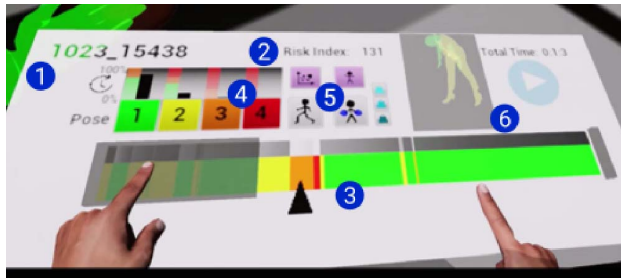


Figure 4: Interactive Analysis panel, providing an overview of the ergonomic evaluation and control over the system: session name (1), Lundqvist index (2), timeline (3), pose class filters (4), heatmap filters (5), replay and preview (6)

basic information (name and duration) and gives an overview of the loaded evaluation session by displaying the OWAS risk index and a timeline of the ergonomic results. This provides the user with an insight into the overall ergonomic quality of the session.

2.4.2 Timeline

The timeline on the analysis panel shown in Figure 4 gives an overview of the ergonomic quality over the course of an evaluation session and shows how often and how long an ergonomically risky position occurred. Colors indicate posture classes from green / low risk to red / highest risk. To support the time-related risk evaluation, the timeline also visualizes how the time share of each pose evolves over time, using a small gray-scale stripe on top of the timeline. The final time share of each pose category is visualized on the upper left of the analysis panel as a gray bar for each pose. An additional colored bar helps to identify the risk level according to the time-related risk evaluation (e.g., high risk when posture class two has a time share of $> 70\%$).

2.4.3 Pose Class Filters

Since it is important to reduce high-risk poses, users need a way to find problematic areas spatially. Our system provides a pose-based filter that gives users information about where a certain pose class occurred, and what the pose actually looks like. As shown in Figure 1, the pose is visualized by the avatar, colored accordingly with the OWAS grades for each body region. The carried weight is visualized by two spheres on the avatar's hands that are colored accordingly. A colored circle on the floor surrounding the pose avatar represents the OWAS class over all body regions.

To offer meaningful filters, not all postures of a certain type can be visualized as this would crowd the scene and overwhelm the user. Therefore, one representative pose is chosen for each separate pose category sequence on the timeline. We found that the pose in the temporal middle of the sequence is most fitting as a representation of the whole sequence. To avoid clutter, poses are only created if no other pose of the same class already exists close to its timestamp.

To link a pose representation with the corresponding sequence in the timeline, an additional timeline is displayed over the pose avatar (Figure 1 left) when users look directly at it.

2.4.4 Motion Trails

To give users a spatial overview of the motion and to identify important areas of different types of user movement, the system offers a filterable heatmap to show the spatial occupancy of user movement (*motion trails*). A 'walking' filter shows areas where nothing was carried, while a 'carrying' filter lets users filter the heatmap to show only movement based on one of three weight grades of carried loads. Figure 5 shows motions trails for two different carried loads.

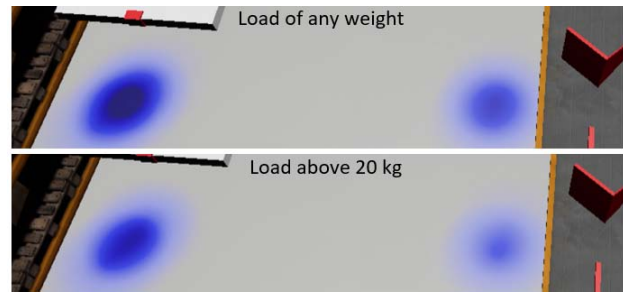


Figure 5: Motion trails showing where the user carried objects, either of any weight (top) or filtered to specific a weight class (bottom). Lighter blues indicate lower occupancy.

2.4.5 Replay

For in-depth details on specific pose sequences, our system offers a replay functionality, enabling users to experience certain time frames or even the whole evaluation session with no gaps or loss of information. The replay button to the upper right of the analysis panel starts and pauses the replay. An additional dynamic preview of the colored avatar is displayed next to the play button when moving the time indicator (see Figure 4).

3 EXPERT INTERVIEW

To evaluate the feasibility of the presented system, we consulted an expert with 30 years of professional experience in physiotherapy. As an expert on motion analysis, judging the quality of motion is one of the primary tasks in her daily work. The main topics discussed in the interview were (1) ergonomics and its validation methods, (2) the applicability of our virtual avatar, and (3) the usefulness of immersive analytics to assess the ergonomics of motion.

3.1 Ergonomics Validation

According to the expert, many physiotherapists or occupational therapist rely more on their own observations and judgment than on any standardized methods. Still, OWAS is a frequently used method internationally, and although it is a rather coarse ergonomics evaluation method, the physiotherapist found it sensible to use a standardized method to assess the quality of motion. Even though it cannot fully substitute the assessment of an expert, it can help to identify problematic or potentially harmful poses or movements.

The physiotherapist also highlighted a potential limitation of our tracking method together with the OWAS method: the bending of spine and neck cannot be fully tracked without an additional tracker around the sternum. However, after reviewing a side-by-side video of the virtual avatar and a real user, she noted that the avatar approximates the real user motion surprisingly closely, despite the missing tracker. While this is encouraging, we feel using an additional tracker would still be worthwhile.

Another positive aspect of our analysis method, recognized by the physiotherapist, is the fact that we measure the time a person remains in a certain pose, since performing similar motions becomes more problematic and potentially harmful when performed over longer periods of time.

3.2 Applicability of the Virtual Avatar

We asked the physiotherapist to judge the usefulness of our virtual avatar, whether she thought it was a suitable representation of a person's motion, which body parts are the most important to track and if using such a tracking system would be valuable in practice.

The physiotherapist pointed out that the spine, the shoulders, and the knees are the most important body parts for an ergonomic assessment, thus the usefulness of our avatar for this assessment depends

on the tracking accuracy of these body parts. After comparing videos of avatar and real user, the physiotherapist observed that the angular position of the knees when bending the legs did not match perfectly. However, she also found this to be an acceptable inaccuracy with no practical relevance for the ergonomics evaluation.

Setting up the whole tracking system for physiotherapy might not be practicable, according to the physiotherapist. However, she mentioned that for occupational therapists, it would be valuable to evaluate whether people can perform certain movements that are typical in their daily work routine and are therefore ready to go back to work or not. A VR setting allows to simulate different work setups that can otherwise not be recreated without a significant effort for a therapy session. Even though it might not be appropriate to use a whole-body VR setup in everyday therapy, the expert deemed the system useful to “test whether a therapy was successful” in the end.

3.3 Usefulness of Immersive Analytics

We also asked which benefits the physiotherapist saw in the immersive analysis of a user’s movement based on a 3D avatar. In her work, a 3D view of the body is essential to get a holistic view of a person’s motion. Her current workflow to assess quality of motion, however, only employs classic videos. Hence, a 3D representation of a person that can be viewed from arbitrary angles (in VR) would improve the process, in particular for remote therapy during a pandemic. She also found that the immersiveness of VR supports learning the correct motion for a task. “People learn through observation. The more holistically I can observe something, the better I can learn what to do.” Our physiotherapist said that she would like to use our system to let people perform the same movements over longer periods of time and evaluate the strain it has on the body. She also mentioned that the system could be very useful for non-expert users, because it allows them to identify unhealthy movements or poses as a problem.

4 CONCLUSION & FUTURE WORK

We presented an immersive analytics system that demonstrates the potential of immersive technologies for ergonomics analysis. Our system provides spatial and temporal overviews of the evaluation session and offers tools to immersively investigate the ergonomic assessment. It allows analyzing in the same VE the data was captured in, either in real-time or after motion tracking and evaluation. In addition, an expert interview with a physiotherapist confirmed the usefulness and sufficient accuracy of our approach. To answer research question *Q1* (Section 1), we developed an ergonomics evaluation method based on a virtual avatar simulation and OWAS, whose value was acknowledged by in the interview. We address *Q2* with a prototypical immersive analytics system (Section 2), supported by the qualitative feedback of the interview that there is potential in capturing and analyzing ergonomic aspects in the same VE.

The current prototype still has some limitations, which should be addressed in future work. Using HTC Vive trackers is an affordable solution for motion tracking, but trackers might shift slightly when users are changing direction quickly. We suggest using a sixth tracker mounted on the sternum, which could further improve the accuracy of the system. Currently, our prototype only supports grasping with one hand. More complex and physical weight-based grabbing and bimanual lifting could further increase fidelity and realism. Still, VR is not yet able to perfectly replicate the behavior of users in the real world [16, 17], and our approach does not solve this problem yet. However, the implemented ergonomic method and visualization can easily be adapted to work with more sophisticated motion tracking systems and setups with physically tangible co-located objects. This does not yet answer *Q3*, but lays the foundation for a system capable of comparing different ergonomics evaluation methods and tracking systems in the future.

An interface to add and modify ergonomics evaluation methods would also be important for experts who have their own, more

specialized or patient-based evaluation methods. We also plan to perform a quantitative evaluation of the accuracy of our proposed avatar and further refine the usability of our immersive ergonomics analysis system by conducting a study with experts and novice users.

Despite the mentioned limitations, we believe that our approach and the developed system constitutes a promising basis for further work on immersive analytics for ergonomics evaluation in VR.

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